

Gursimran Panesar

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Mathematics (Computer Science & Mathematics)

May 2026

Pittsburgh, PA

EXPERIENCE

Software Engineer

Aug. 2026 – Present

Doordash

- New Verticals + Doordash AI Research Labs

Software Engineering Intern

May 2026 – Aug. 2026

Meta

- Migrated 6 Instagram endpoints from Django to Hack, decreasing infrastructure latency for 100M–1B+ users
- Generated \$12M in annual infrastructure savings through backend migration resource utilization improvements
- Optimized ranking for follow recommendations to increase follows/reciprocal follows by 2% for daily active users
- Created and integrated agentic development harness, reducing required human intervention by 70% on migrations

AI Safety Researcher

Jan. 2026 – Present

Carnegie Mellon University

- Researched AI behavior under long-context and agentic scenarios, benchmarking 10+ state-of-the-art models
- First author on paper targeting ICLR 2027: monitoring methods for detecting unsafe behavior in LLM/agent trace

Software Engineering Intern

May 2025 – Aug. 2025

Amazon Web Services (AWS)

- Integrated 10+ diagnostic capabilities into CloudWatch console to reduce context switching during investigation
- Built MCP server with AI retrieval and tool orchestration to accelerate customer-directed investigations by 60%
- Reduced navigation away from core console by 10%, improving efficiency by keeping users within primary workflow

Software Engineering Intern, AI & Automation

May 2024 – Aug. 2024

Bank of New York (BNY)

- Implemented large language model agent using PyTorch to read legal loan documents, saving 300hrs per quarter
- Deployed Azure character recognition model for W9 forms to increase processing speed into SQL database by 97%
- Researched AI use-cases and presented 5-year plan to executives for safe implementation of AI/ML technologies

AI Robotics Researcher

Dec. 2023 – Aug. 2024

Carnegie Mellon AirLab

- Programmed critical user mapping features for drone path planning to elevate experience and efficiency using C++
- Created Node.js dashboard for 100 drones to monitor 50 health indicators transmitting from vehicle to base station

PROJECT EXPERIENCE

AI Sycophancy Evaluation and Mitigation

- Investigated sycophancy in Gemma-4 model through mechanistic interpretability (under Prof. Aran Nayebi)
- Presented mitigation through Supervised Fine-Tuning, Low-Rank Adaptation, Direct Preference Optimization

Microsoft Tiny Troupe Agentic AI

- Collaborated with Microsoft Research in evaluating Tiny Troupe framework for agentic behavioral economics
- Benchmarked Tiny Troupe for behavioral economics experiments to improve ability to replicate human behavior

ML x Twitter Stock Market

- Developed logistic regression model using Numpy and Pandas to determine sentiment and predict stock trajectory
- Achieved 93% accuracy rate in predicting short term trends of stock prices based on dataset of 184k tweets

SKILLS & CERTIFICATIONS

Certifications: Amazon AWS Cloud Practitioner, Microsoft Azure AI Fundamentals, IBM Watson AI Fundamentals

Programming: Python, C, Standard ML, JavaScript, Java, C++, SQL, ROS, Unix/Linux, Node.js

Machine Learning: PyTorch, TensorFlow, Sklearn, Numpy, Pandas, Jupyter, LLMs, Model Context Protocol

Coursework (Grad/PhD): Generative AI, Deep Learning, Natural Language Processing, Machine Learning